



Cambridge International AS & A Level

PHYSICS

9702/41

Paper 4 A Level Structured Questions

October/November 2021

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

- 1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
- 2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
- 3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
- 4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
- 5 'List rule' guidance
 For questions that require ***n*** responses (e.g. State **two** reasons ...):
 - The response should be read as continuous prose, even when numbered answer spaces are provided.
 - Any response marked *ignore* in the mark scheme should not count towards ***n***.
 - Incorrect responses should not be awarded credit but will still count towards ***n***.
 - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
 - Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Abbreviations

/	Alternative and acceptable answers for the same marking point.
()	Bracketed content indicates words which do not need to be explicitly seen to gain credit but which indicate the context for an answer. The context does not need to be seen but if a context is given that is incorrect then the mark should not be awarded.
—	Underlined content must be present in answer to award the mark. This means either the exact word or another word that has the same technical meaning.

Mark categories

B marks	These are <u>independent</u> marks, which do not depend on other marks. For a B mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer.
M marks	These are <u>method</u> marks upon which A marks later depend. For an M mark to be awarded, the point to which it refers must be seen specifically in the candidate's answer. If a candidate is not awarded an M mark, then the later A mark cannot be awarded either.
C marks	These are <u>compensatory</u> marks which can be awarded even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known them. For example, if an equation carries a C mark and the candidate does not write down the actual equation but does correct working which shows the candidate knew the equation, then the C mark is awarded. If a correct answer is given to a numerical question, all of the preceding C marks are awarded automatically. It is only necessary to consider each of the C marks in turn when the numerical answer is not correct.
A marks	These are <u>answer</u> marks. They may depend on an M mark or allow a C mark to be awarded by implication.

Annotations

✓	Indicates the point at which a mark has been awarded.
X	Indicates an incorrect answer or a point at which a decision is made not to award a mark.
XP	Indicates a physically incorrect equation ('incorrect physics'). No credit is given for substitution, or subsequent arithmetic, in a physically incorrect equation.

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ECF	Indicates 'error carried forward'. Answers to later numerical questions can always be awarded up to full credit provided they are consistent with earlier incorrect answers. <u>Within</u> a section of a numerical question, ECF can be given after AE, TE and POT errors, but not after XP.
AE	Indicates an arithmetic error. Do not allow the mark where the error occurs. Then follow through the working/calculation giving full subsequent ECF if there are no further errors.
POT	Indicates a power of ten error. Do not allow the mark where the error occurs. Then follow through the working/calculation giving full subsequent ECF if there are no further errors.
TE	Indicates incorrect transcription of the correct data from the question, a graph, data sheet or a previous answer. For example, the value of 1.6×10^{-19} has been written down as 6.1×10^{-19} or 1.6×10^{19} . Do not allow the mark where the error occurs. Then follow through the working/calculation giving full subsequent ECF if there are no further errors.
SF	Indicates that the correct answer is seen in the working but the final answer is incorrect as it is expressed to too few significant figures.
BOD	Indicates that a mark is awarded where the candidate provides an answer that is not totally satisfactory, but the examiner feels that sufficient work has been done ('benefit of doubt').
CON	Indicates that a response is contradictory.
I	Indicates parts of a response that have been seen but disregarded as irrelevant.
M0	Indicates where an A category mark has not been awarded due to the M category mark upon which it depends not having previously been awarded.
^	Indicates where more is needed for a mark to be awarded (what is written is not wrong, but not enough). May also be used to annotate a response space that has been left completely blank.
SEEN	Indicates that a page has been seen.

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Question	Answer	Marks
1(a)	constant speed or constant magnitude of velocity	B1
	acceleration (always) perpendicular to velocity	B1
1(b)(i)	$F = mv^2 / r$ or $v = r\omega$ and $F = mr\omega^2$	C1
	$F = 790 \times 94^2 / 318$ $= 22\,000 \text{ N}$	A1
1(b)(ii)	centripetal acceleration: same	B1
	maximum speed: greater	B1
	time taken for one lap of the track: greater	B1

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Question	Answer	Marks
2(a)	work done per unit mass	B1
	(work done in) moving mass from infinity	B1
2(b)(i)	(gravitational) fields from the Earth and Moon are in opposite directions	B1
	(resultant is zero where gravitational) fields are equal (in magnitude)	B1
2(b)(ii)	$g \propto M / r^2$	C1
	$5.98 \times 10^{24} / x^2 = 7.35 \times 10^{22} / (3.84 \times 10^8 - x)^2$ leading to $x = 3.5 \times 10^8$ (m)	A1
2(b)(iii)	ϕ (Earth) = $(-6.67 \times 10^{-11} \times (5.98 \times 10^{24} / 3.5 \times 10^8))$ and ϕ (Moon) = $(-6.67 \times 10^{-11} \times (7.35 \times 10^{22} / 0.38 \times 10^8))$	C1
	$\phi = (-6.67 \times 10^{-11} \times [(5.98 \times 10^{24} / 3.5 \times 10^8) + (7.35 \times 10^{22} / 0.38 \times 10^8)])$	C1
	$= -1.3 \times 10^6 \text{ J kg}^{-1}$	A1

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Question	Answer	Marks
3(a)	(thermal) energy per unit mass (to cause temperature change)	B1
	(thermal) energy per unit <u>change</u> in temperature	B1
3(b)(i)	$(T =) pV / Nk$	B1
3(b)(ii)	$(pV =) NkT = \frac{1}{3}Nm\langle c^2 \rangle$ or $pV = NkT$ and $pV = \frac{1}{3}Nm\langle c^2 \rangle$	M1
	leading to $\frac{1}{2}m\langle c^2 \rangle = (3/2)kT$ and $\frac{1}{2}m\langle c^2 \rangle = E_K$	A1
3(b)(iii)	internal energy = ΣE_K (of molecules) + ΣE_P (of molecules) or no forces between molecules	B1
	potential energy of molecules is zero	B1
3(c)(i)	increase in internal energy = Q + work done	B1
	constant volume so no work done	B1
3(c)(ii)	$c = Q / Nm\Delta T$	C1
	$= [N \times (3/2)k\Delta T] / (Nm\Delta T) = 3k / 2m$	A1
3(d)	(as it expands) gas does work (against the atmosphere/external pressure)	B1
	for same temperature rise) more (thermal) energy needed, so larger specific heat capacity	B1

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Question	Answer	Marks
4(a)(i)	5.0 cm	A1
4(a)(ii)	$\omega = 2\pi / T$ or $\omega = 2\pi f$ and $f = 1 / T$	C1
	$\omega = 2\pi / 4.0$ $= 1.6 \text{ rad s}^{-1}$	A1
4(a)(iii)	$v_0 = \omega x_0$	C1
	$= 1.57 \times 5.0$ $= 7.9 \text{ cm s}^{-1}$	A1
4(b)	- initial pull was to the right - distance from X to trolley (at equilibrium) is 20 cm - period is 4.0 s - initial motion undamped - motion becomes damped at/from 12 s - damping is light - maximum speed at 1 s, 3 s, etc. / stationary at 2 s, 4 s, etc. <i>Any three points, 1 mark each</i>	B3
4(c)	sketch: closed loop encircling (20, 0)	B1
	minimum L shown as 15 cm <u>and</u> maximum L shown as 25 cm	B1
	minimum v shown as -7.9 cm s^{-1} <u>and</u> maximum v shown as $+7.9 \text{ cm s}^{-1}$	B1

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Question	Answer	Marks
5(a)	- noise can be removed/signal can be regenerated - extra bits can be added for error-checking - signal can be encrypted (for increased security) - data compression/multiplexing is possible <i>Any two points, 1 mark each</i>	B2
5(b)(i)	4 ms: 0101 and 8 ms: 0100	B1
5(b)(ii)	sketch: horizontal line continues to 8 ms, then new horizontal line from 8 ms to 12 ms	B1
	level of line after 8 ms is 4 mV	B1
5(c)	sketch: series of steps of width 2 ms	B1
	step heights at 0, 2, 4, 6, 4, 6 mV <i>2 marks if all correct, 1 mark if only one incorrect</i>	B2

Question	Answer	Marks
6(a)	$Q = CV$ and $E = \frac{1}{2}CV^2$	B1
6(b)(i)	$C_N = CL / (L - D)$	B1
6(b)(ii)	(charge is unchanged by moving the plates so) $Q_N = CV$	B1
6(b)(iii)	$V_N = Q_N / C_N$ $= (CV) / [CL / (L - D)]$ $= V(L - D) / L$	B1
6(c)	oppositely charged plates attract, so energy stored decreases	B1

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Question	Answer	Marks
7(a)	• infinite (open-loop) gain • infinite slew rate • infinite input impedance • zero output impedance • infinite bandwidth <i>Any two points, 1 mark each</i>	B2
7(b)	X: thermistor and Y: relay	B1
7(c)(i)	(any) difference in voltage at the inputs causes output to saturate (because gain is very large)	B1
	saturates positively if $V^+ > V^-$ <u>and</u> saturates negatively if $V^+ < V^-$	B1
7(c)(ii)	comparator	B1
7(c)(iii)	temperature	M1
	above a particular value	A1
7(c)(iv)	to adjust the temperature (at which the lamp illuminates/extinguishes)	B1

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Question	Answer	Marks
8(a)	newton per ampere per metre	M1
	where current/wire is perpendicular to magnetic field	A1
8(b)(i)	$F = BIL \sin \theta$	C1
	$B = 1.0 / (5.0 \times 0.060 \times \sin 50^\circ)$ = 4.4 mT	A1
8(b)(ii)	(from Fleming's left-hand rule) force on wire is upwards, so reading decreases	B1
8(b)(iii)	frame will rotate (so that PQ becomes perpendicular to the field)	B1

Question	Answer	Marks
9(a)	constant voltage	M1
	that produces/dissipates same power as (the mean power of) the alternating voltage	A1
9(b)(i)	(maximum) rate of cutting of (magnetic) flux doubles	B1
	(peak and hence) r.m.s. induced e.m.f. doubles	B1
9(b)(ii)	sketch: (sinusoidal) wave of period 10 ms	B1
	peak E shown as $\pm 34 \text{ V}$ (1 mark out of 2 awarded if peak E shown as $\pm 17 \text{ V}$ or $\pm 24 \text{ V}$)	B2
9(c)	current in the coil results in forces that oppose its rotation or current in the resistor dissipates the energy of rotation	B1
	coil stops rotating	B1

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Question	Answer	Marks
10(a)(i)	photoelectric effect	B1
10(a)(ii)	electron diffraction	B1
10(b)(i)	$\lambda = h / p$	M1
	h is the Planck constant	A1
10(b)(ii)	de Broglie (wavelength)	B1
10(c)(i)	$\frac{1}{2}mv^2 = eV$	C1
	$\frac{1}{2} \times 9.11 \times 10^{-31} \times v^2 = 1.60 \times 10^{-19} \times 4800$ <u>so</u> $v = 4.1 \times 10^7 \text{ m s}^{-1}$	A1
10(c)(ii)	$\lambda = h / mv$ $= 6.63 \times 10^{-34} / (9.11 \times 10^{-31} \times 4.1 \times 10^7)$	C1
	$= 1.8 \times 10^{-11} \text{ m}$	A1

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Question	Answer	Marks
11(a)(i)	ease with which edges can be distinguished	B1
11(a)(ii)	difference in degrees of blackening	B1
11(b)	$I = I_0 \exp(-\mu x)$	C1
	$0.12 = \exp(-\mu \times 2.3)$	C1
	$\ln 0.12 = -2.3 \times \mu$	
	$\mu = 0.92 \text{ cm}^{-1}$	A1
11(c)	advantage: produces 3-dimensional image	B1
	disadvantage: (much) greater exposure to radiation	B1

Question	Answer	Marks
12(a)	probability of decay (of a nucleus)	M1
	per unit time	A1
12(b)	$A = \lambda N$	C1
	$N = \text{mass} / (\text{nucleon number} \times u)$	C1
	$2.92 \times 10^9 = (\lambda \times 5.87 \times 10^{-10}) / (131 \times 1.66 \times 10^{-27})$	A1
	$\lambda = 1.08 \times 10^{-6} \text{ s}^{-1}$	
12(c)	- sample emits radiation in all directions - some radiation is absorbed by air/detector window - self-absorption within the source - dead time/inefficiency of detector Any two points, 1 mark each	B2